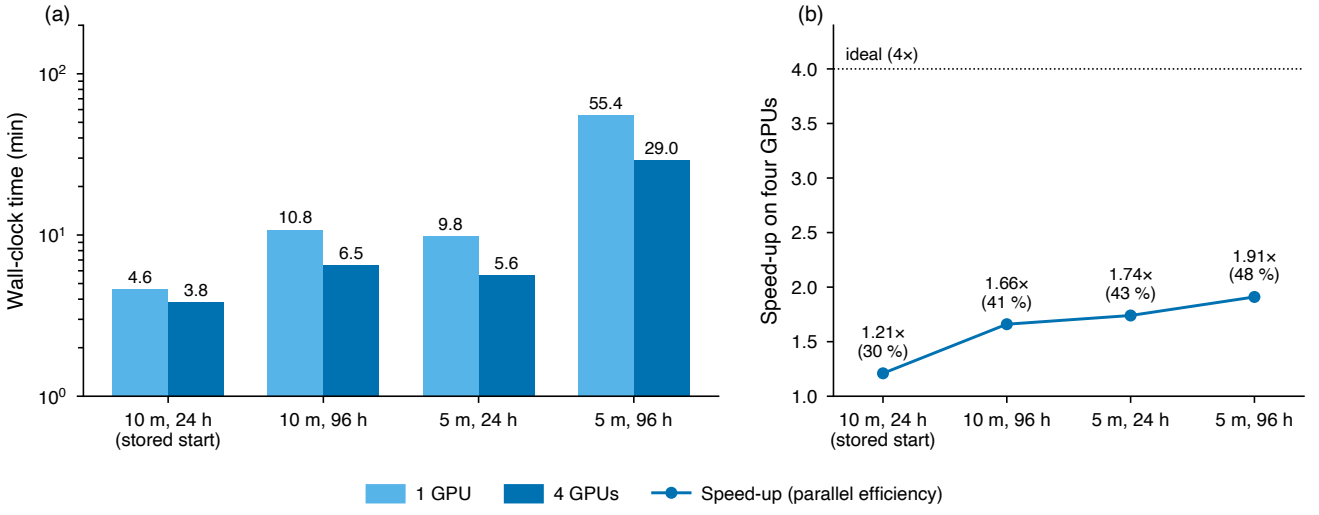
(a)
Wall-clock time (min)
$10^2$
$10^1$
$10^0$
4.6
3.8
10.8
6.5
9.8
5.6
55.4
29.0
10 m, 24 h
(stored start)
10 m, 96 h
5 m, 24 h
5 m, 96 h
(b)
Speed-up on four GPUs
ideal (4×)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.21×
(30 %)
1.66×
(41 %)
1.74×
(43 %)
1.91×
(48 %)
10 m, 24 h
(stored start)
10 m, 96 h
5 m, 24 h
5 m, 96 h
1 GPU
4 GPUs
Speed-up (parallel efficiency)